

Semester	B.E. Semester VII – Computer Engineering
Subject	R Programming
Subject Professor In- charge	Prof. Sanjeev Dwivedi
Assisting Teachers	Prof. Sanjeev Dwivedi
Laboratory	M-411B

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Roll Number	23102B0055
Grade and Subject Teacher's Signature	

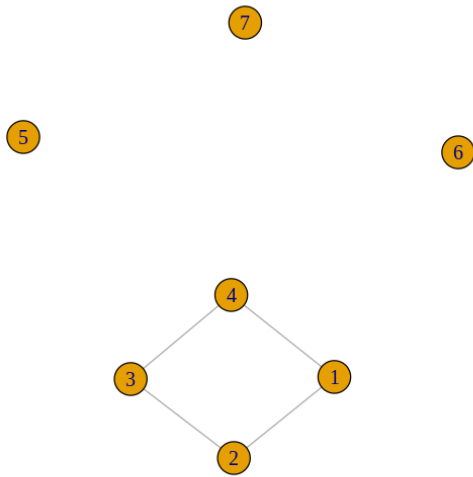
Experiment Number	05
Experiment Title	Social Network Analysis with R

Resources / Apparatus Required	Hardware: Computer system	Software: Colab
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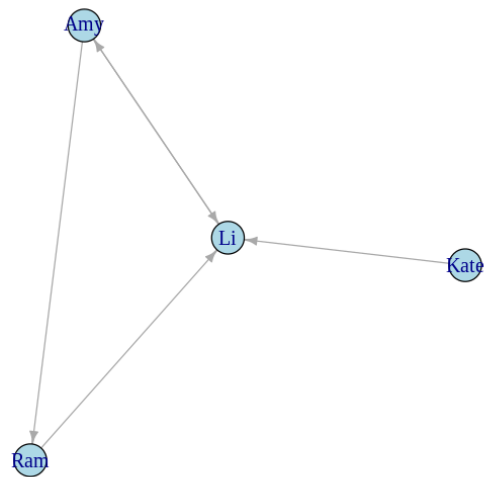
Objective	Study relationships between entities using R, model them as node-edge network, apply igraph to compute structural measures, find influential nodes, detect communities.
Description	Notebook build network step-by-step: • Setup: install/load igraph. • Toy graphs: small undirected + directed graph show node/edge representation basics. • Toy measures: degree (in/out/all), diameter, density, reciprocity, closeness, betweenness, transitivity, assortativity computed on toy graph g1. • Real dataset: edge list pulled from GitHub CSV, built into directed graph net via graph.data.frame. • Degree distribution: histogram + summary stats show connection spread. • Network measures: diameter, density, reciprocity, closeness, betweenness, transitivity, assortativity computed on real network. • Visualization: raw plot, degree-sized/colored plot, heatmap style plot (darker = more connected). • Hub/Authority scores: hub_score() find connectors, authority.score() find popular/pointed-to nodes. • Centrality table: combine degree, in/out-degree, closeness, betweenness, hub, authority into ranked table; top hubs/authorities printed. • Community detection: edge-betweenness clustering + Louvain clustering, modularity compared between two methods. • Outputs saved: centrality CSV + plot PNGs to outputs/ folder for submission.

Output

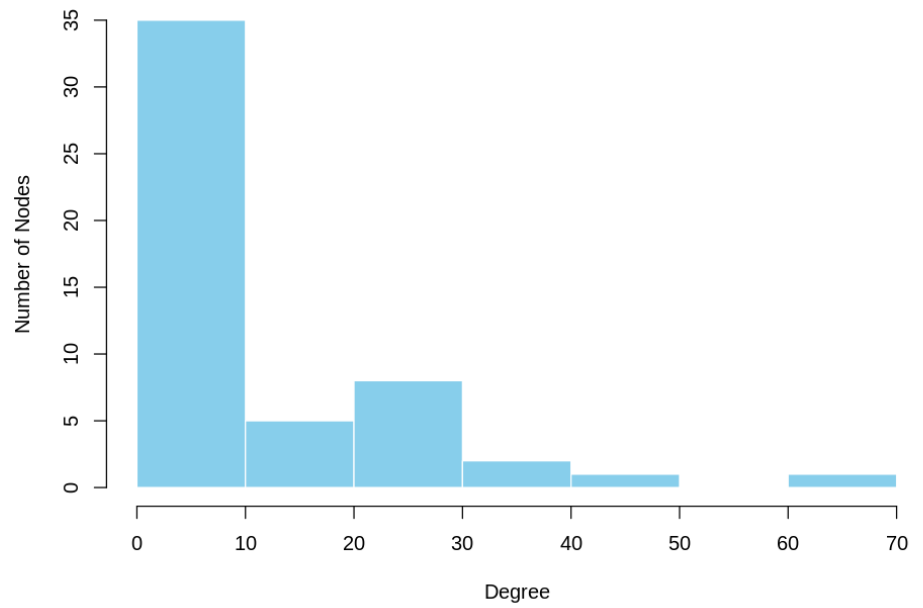
Toy Undirected Graph



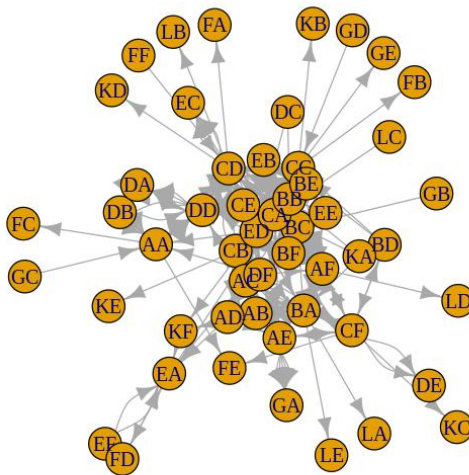
Toy Directed Social Network



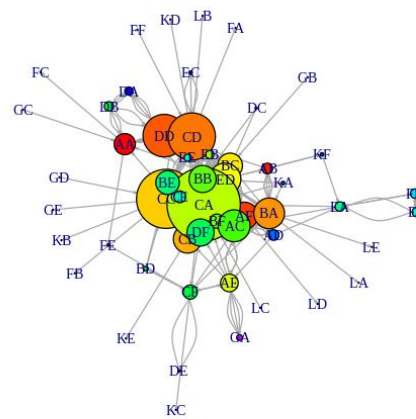
Distribution of Node Degree



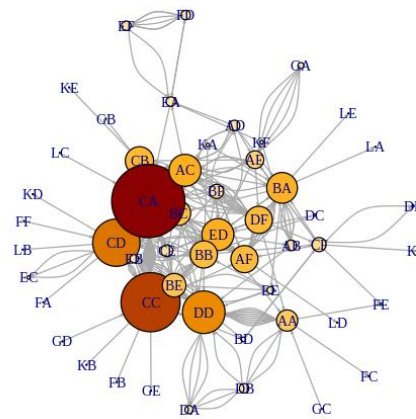
Raw Network Diagram



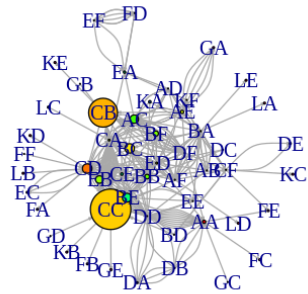
Network Sized & Colored by Degree



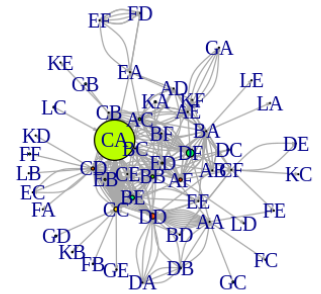
Degree Heat-map (darker = more connections)



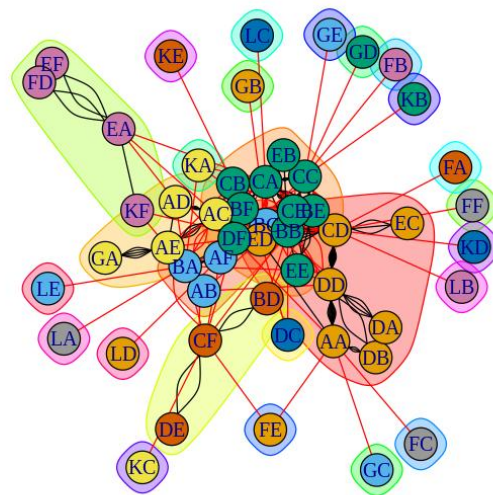
Hubs

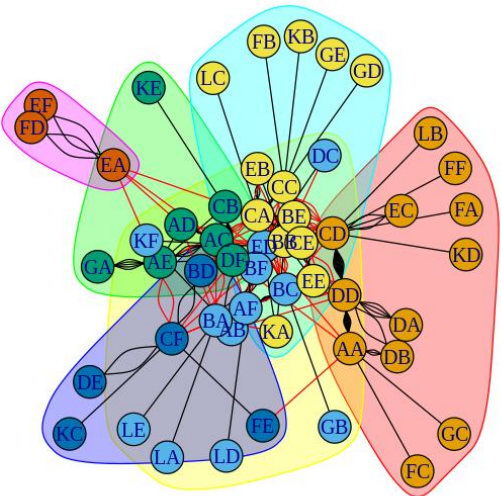


Authorities



Community Detection (Edge-Betweenness)



	Community Detection (Louvain) 
Conclusion	Network show hub-and-spoke degree pattern, few high-degree nodes act as key connectors/brokers. Centrality and hub/authority scores identify most influential nodes. Community detection reveal distinct sub-groups, Louvain and edge-betweenness mostly agree on structure, confirm igraph effective for social network analysis.